Sarv Pariksha

परीक्षा सबकी, तैयारी पक्की · sarvpariksha.in

User Guide — Visitors & Guests

Everything you can do on sarvpariksha.in **without creating an account** — practise real papers in a genuine exam interface, free.

1. What is SarvPariksha?

SarvPariksha (*सर्व परीक्षा* — "every exam") is a free exam-practice platform built for two audiences: **students** who want to attempt previous-year papers, official NTA mock tests and AI-generated challenge exams in a real computer-based-test (CBT) interface, and **teachers** who want to launch a shareable, auto-graded exam in minutes instead of hand-building a Google Form.

NEW Now you can create your own practice papers! With Instant Share feature!

Practice on real previous-year papers

101+ mock exams across 7 categories. AI hints when you're stuck, full explanations after every question. Free, no signup required.

Start practicing → Browse exams

SAMPLE QUESTION Q 1/3

Which planet is known as the Red Planet?

A Venus
B Mars ✓
C Jupiter
D Saturn

AI EXPLANATION
Mars appears red because of iron oxide (rust) on its surface, so it is called the Red Planet.

Exams

View all →

JEE Mains	Physics, Chemistry & Math	75 Qs • 3 hrs
JEE Advanced	Physics, Chemistry & Math	75 Qs • 3 hrs
NEET	Biology, Chemistry & Physics	200 Qs • 3.5 hrs
GATE CS	CS, General Aptitude	65 Qs • 3 hrs
MHT-CET	Physics, Chemistry, Math/Bio	150 Qs • 3 hrs
Coding Tests	DSA Logic & Output Prediction	30 Qs • 1 hr
Topic Based	Focus on specific topics & units	Your topics, on demand

What you get

Unlimited mock exams
Full-length tests built on previous-year patterns, free forever. No caps, no trial.

AI hints & explanations
A nudge when you're stuck mid-exam. Full step-by-step explanations after submission. AI can be removed from settings as well.

Instant scoring
Submit and get your score, section-wise breakdown and solutions in seconds.

AI paper predictor
Custom papers generated from the latest exam trends so you're always practicing what matters.

What our users say

"The AI mocks helped me focus on my weak areas. I cracked SSC CGL in my first attempt."
PS **Priya Sharma**
SSC CGL 2023 Qualified

"Topic-wise practice and detailed analytics helped me track progress and improve my speed."
RV **Rahul Verma**
IBPS PO Selected

"I practiced 30 minutes daily on SarvPariksha and it made a huge difference in my preparation."
VP **Vikram Patel**
Railway Group D Selected

Questions, answered

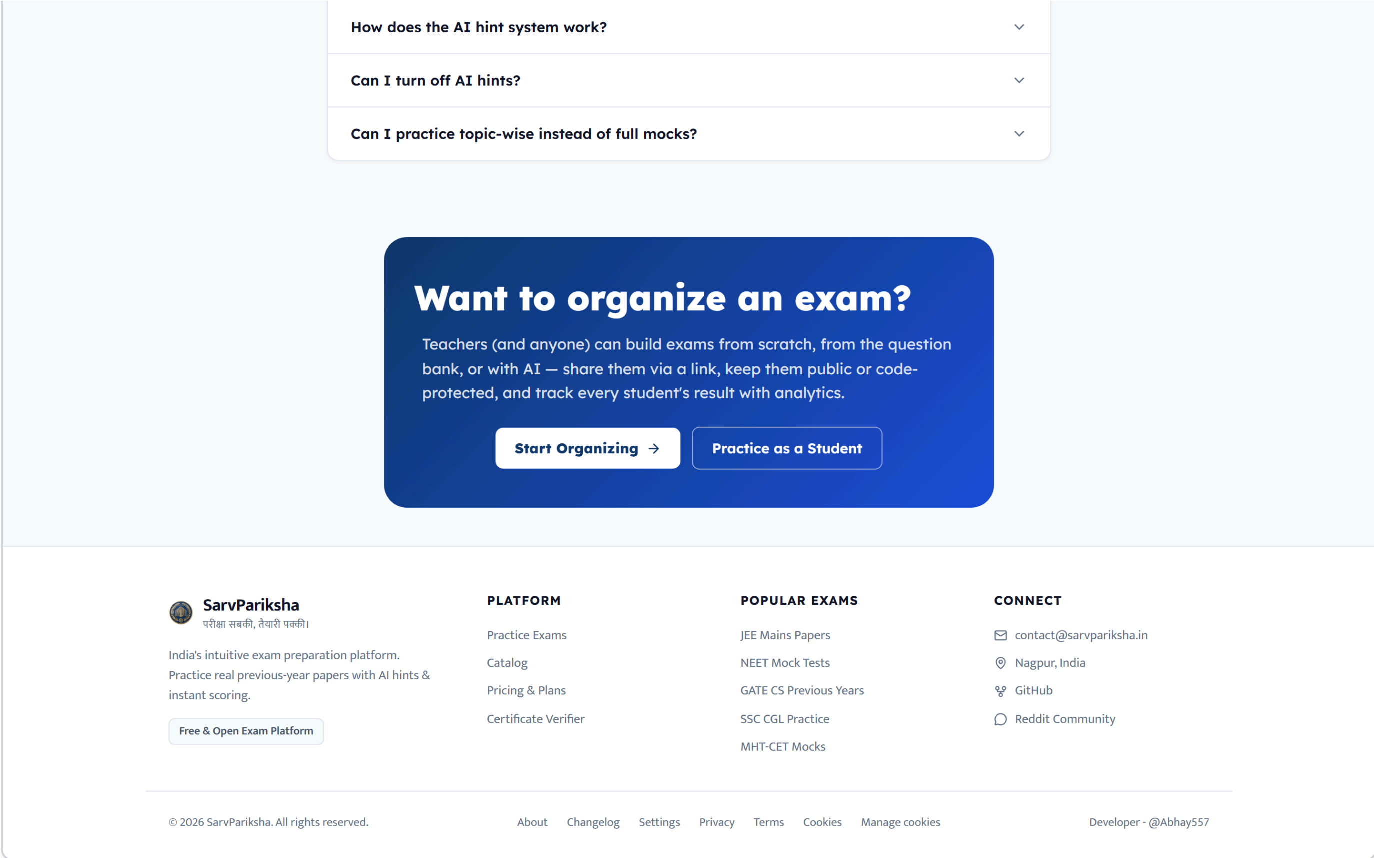
Everything you need to know about SarvPariksha, mock exams, and AI hints.

Is SarvPariksha really free?

Yes. All mock exams, previous-year question papers, scoring and AI explanations are completely free. No credit card, no trial period, just start practicing.

Do I need to create an account to practice?

Which exams are covered?



The SarvPariksha home page

2. What you can do without an account

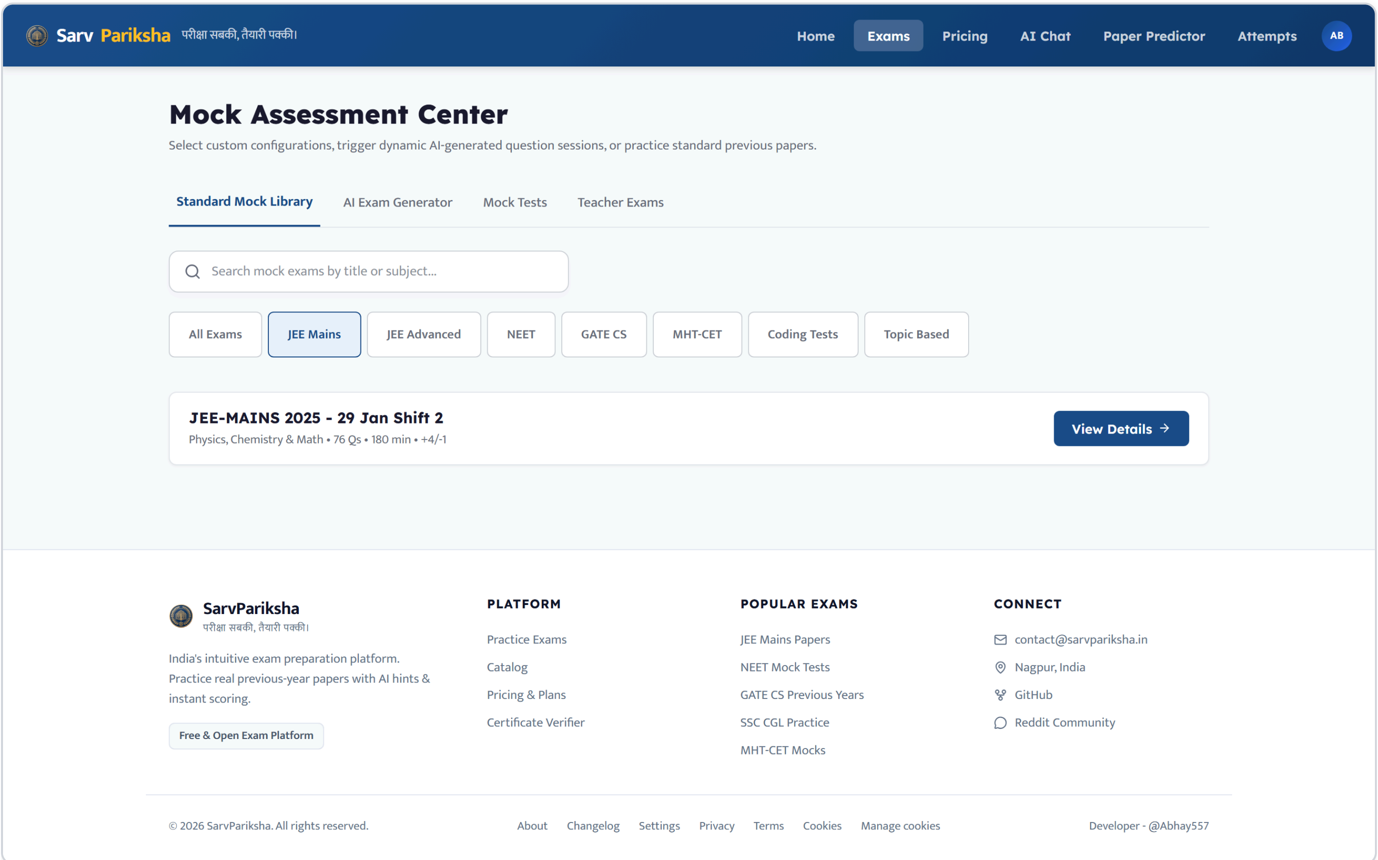
Feature	Where	Notes
Standard Mock Library	Exams → Standard Mock Library	JEE Mains/Advanced, NEET, GATE, MHT-CET, coding & topic-based papers with instant scoring
Official NTA Mock Tests	Exams → Mock Tests	1,700+ real CBT practice papers across 16 exam bodies (UGC-NET, JEE Main, NEET, DUET, SWAYAM...)
Full CBT exam engine	any exam	Sections, question palette, timer or untimed Practice Mode, negative marking, on-screen numeric keypad
Solutions & explanations	after every submit	Every question shows the correct answer with explanation
Verify any certificate	Certificate Verifier	Enter a SARV-serial or scan the QR printed on a certificate
Take a teacher's shared exam	link `/take/TE-XXXXXX`	Private exams need the organizer's code; your score goes to the organizer

> **AI features need a free account.** As a guest you cannot use AI Exam Generation, AI Hints inside exams, AI Chat or the Paper Predictor — those unlock the moment you create a free account.

> **Your data stays yours.** As a guest, everything — every paper you open, every answer, every score — is saved **only in this browser's local storage**. Nothing is transmitted to or stored on SarvPariksha's servers. The database only becomes involved when you sign in (to sync history) or use multi-user features like teacher exams. Clearing site data clears guest attempts.

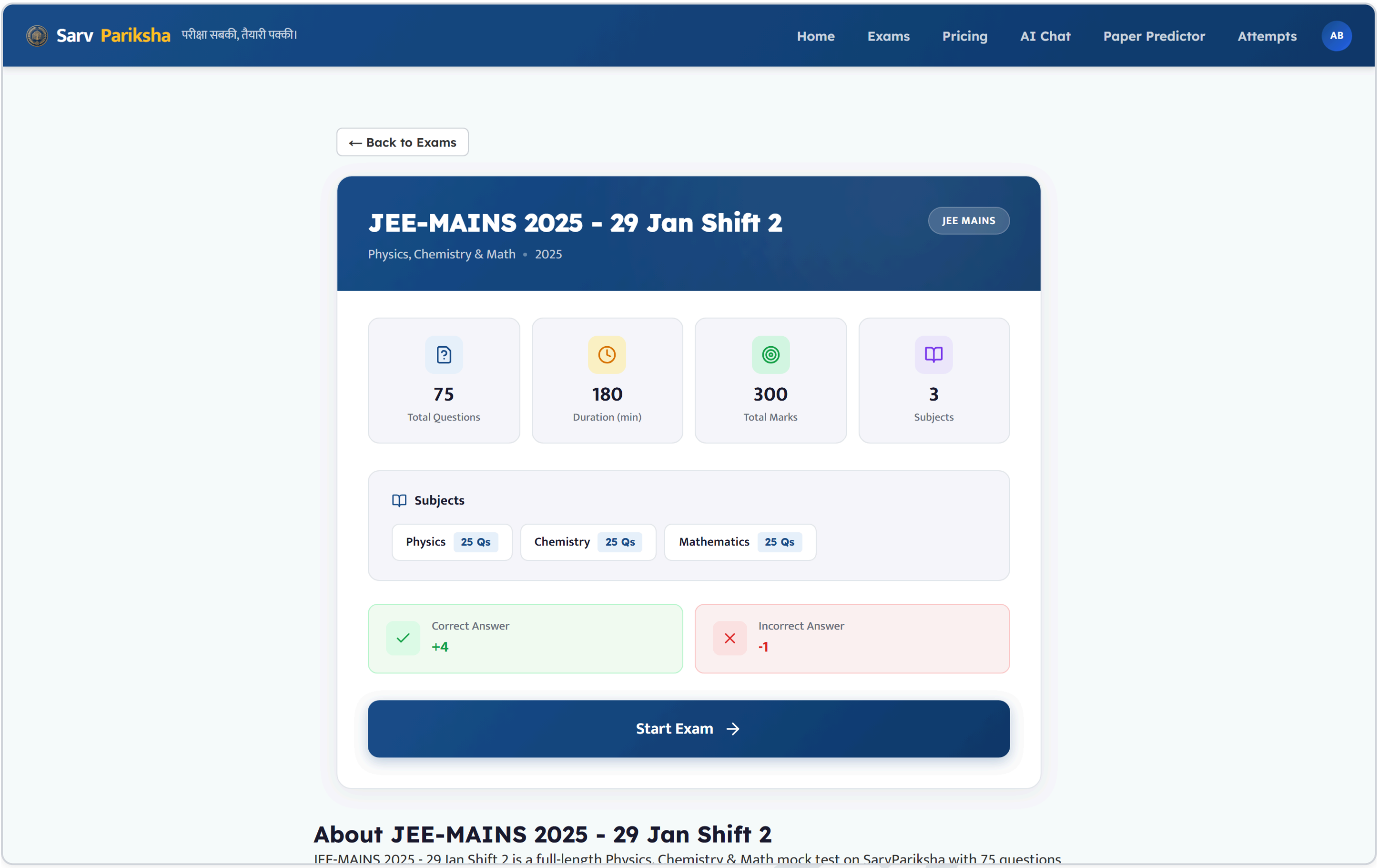
3. Take your first mock exam

1. Open **Exams** in the top navigation. The *Standard Mock Library* lists every bundled paper — search or filter by category.



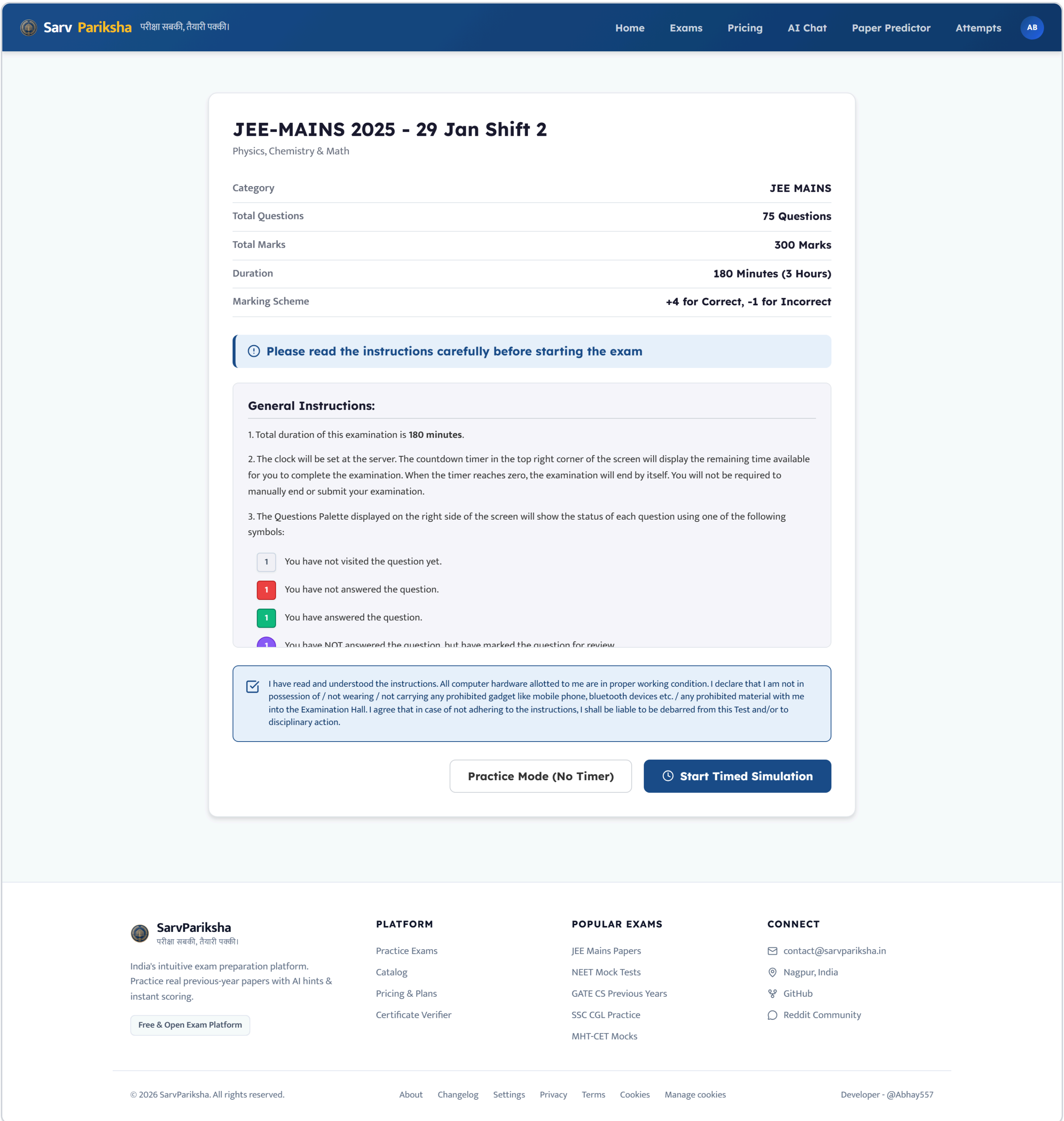
Standard Mock Library — search, category chips, one-click paper list

2. Click **View Details** on any paper to see its stats — questions, duration, total marks and marking scheme.



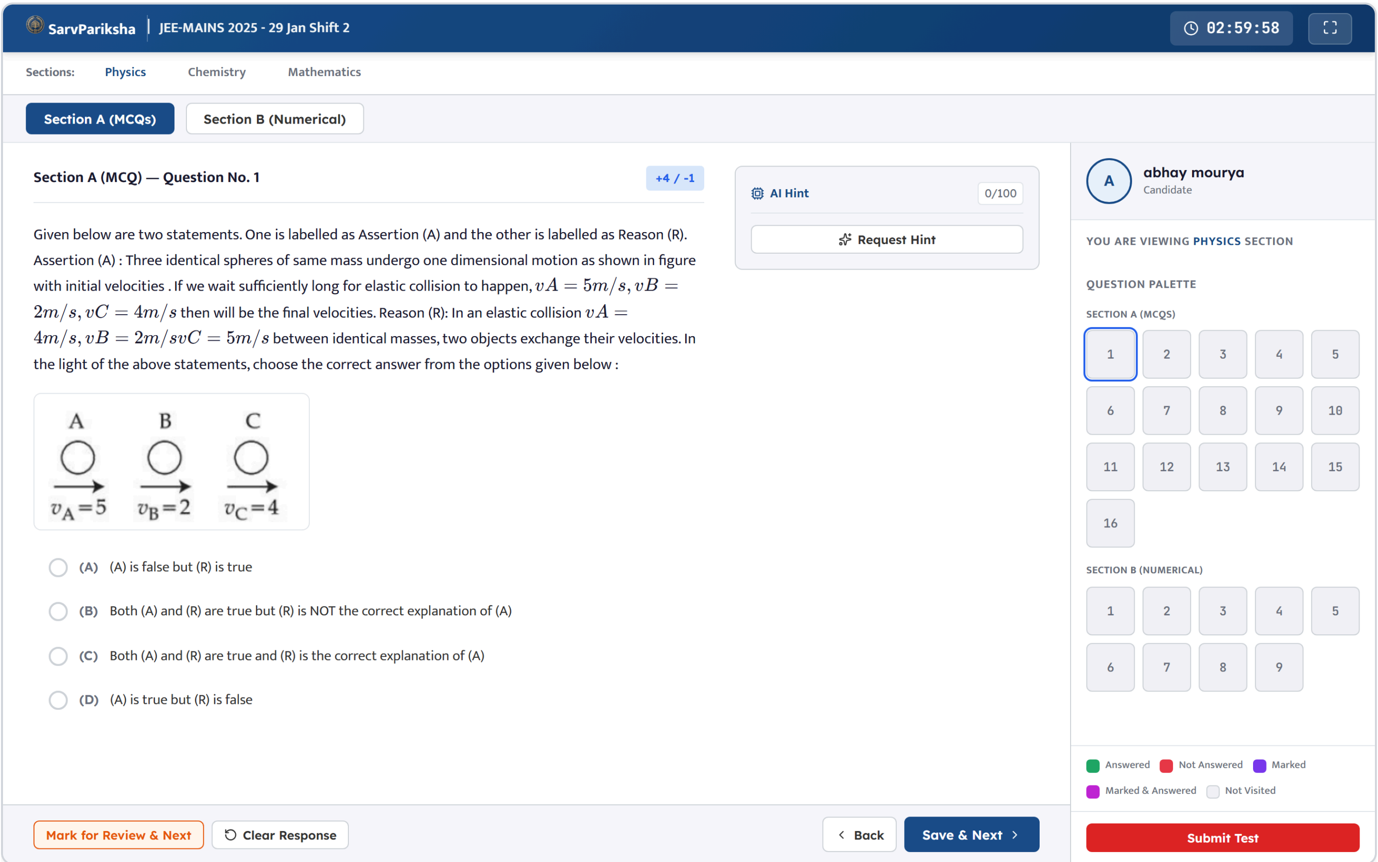
Exam detail — stats, subjects and marking before you commit

3. Press **Start Exam**. You'll see an NTA-style instruction page. Read the instructions and tick the declaration checkbox — this mirrors the real exam-day flow.



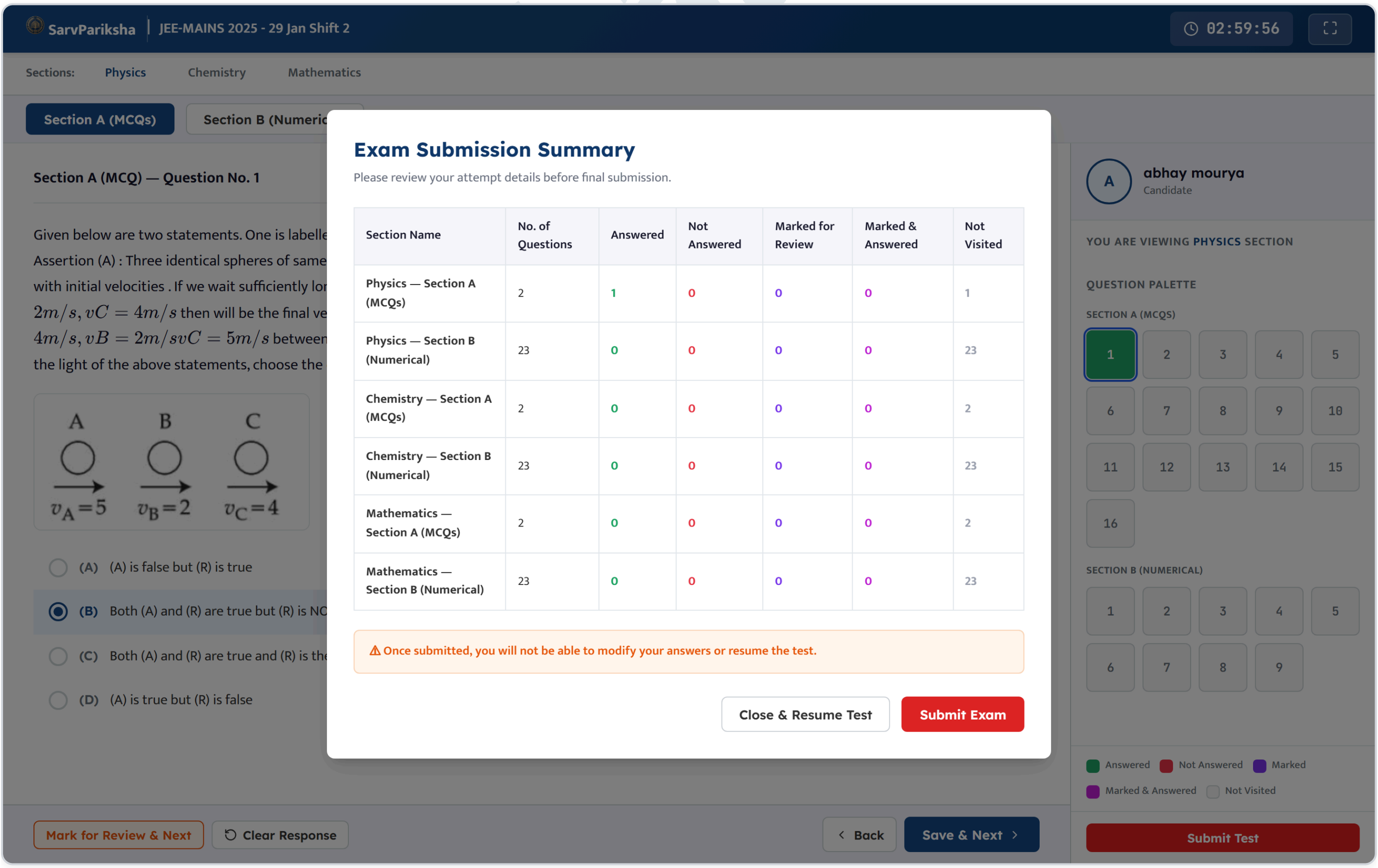
Pre-start instructions with the mandatory declaration

4. Choose **Start Timed Simulation** (real exam feel) or **Practice Mode** (no timer). Inside the exam: pick answers, use the **question palette** to jump around, **Mark for Review** to flag doubts, and the on-screen keypad for numerical questions.



The CBT engine — sections, palette, timer, AI hint panel

5. Press **Submit Test** and confirm. Your scorecard appears instantly with correct/incorrect/skipped breakdown, followed by full solutions for every question.



Submit confirmation — shows an answered/unanswered summary

Performance Scorecard

JEE-MAINS 2025 - 29 Jan Shift 2

0%

-1 / 300

Correct	0
Incorrect	1
Skipped	74
Time Taken	0m 4s
Total Marks	300

Standard Practice Mock Exam

This is a standard free practice mock exam. Official cryptographically verifiable accomplishment certificates are only awarded for custom AI-Generated exams scoring ≥ 40%.

Solutions & Explanations

Physics — Section A — Q1.

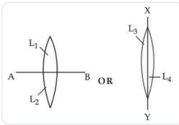
Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A) : Three identical spheres of same mass undergo one dimensional motion as shown in figure with initial velocities . If we wait sufficiently long for elastic collision to happen, $vA = 5m/s, vB = 2m/s, vC = 4m/s$ then will be the final velocities. Reason (R): In an elastic collision $vA = 4m/s, vB = 2m/svC = 5m/s$ between identical masses, two objects exchange their velocities. In the light of the above statements, choose the correct answer from the options given below :

X Wrong (B)

Correct Answer: A

Physics — Section A — Q2.

Two identical symmetric double convex lenses of focal length f are cut into two equal parts L_1, L_2 by AB plane and L_3, L_4 by XY plane as shown in figure respectively. The ratio of focal lengths of lenses L_1 and L_3 is



— Skipped

Correct Answer: B

Physics — Section B — Q1.

Two bodies A and B of equal mass are suspended from two massless springs of spring constant k_1 and k_2, respectively. If the bodies oscillate vertically such that their amplitudes are equal, the ratio of the maximum velocity of A to the maximum velocity of B is

— Skipped

Correct Answer: B

Physics — Section B — Q2.

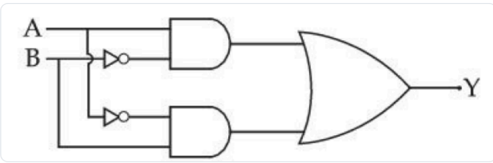
Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A) : With the increase in the pressure of an ideal gas, the volume falls off more rapidly in an isothermal process in comparison to the adiabatic process. Reason (R) : In isothermal process, constant, $PV = PV\gamma =$ while in adiabatic process constant. Here is the ratio of specific heats, P is the pressure and V is the volume of the ideal gas. In the light of the above statements, choose the correct answer from the options given below :

— Skipped

Correct Answer: A

Physics — Section B — Q3.

The truth table for the circuit given below is :



— Skipped

Correct Answer: D

Physics — Section B — Q4.

The difference of temperature in a material can convert heat energy into electrical energy. To harvest the heat energy, the material should have

— Skipped

Correct Answer: D

Physics — Section B — Q5.
Match List - I with List - II. Choose the correct answer from the options given below :

	List - I		List - II
(A)	Young's Modulus	(I)	$M L^{-1} T^{-1}$
(B)	Torque	(II)	$M L^{-1} T^{-2}$
(C)	Coefficient of Viscosity	(III)	$M^{-1} L^3 T^{-2}$
(D)	Gravitational Constant	(IV)	$M L^2 T^{-2}$

— Skipped

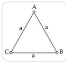
Correct Answer: C

Physics — Section B — Q6.
A sand dropper drops sand of mass $m(t)$ on a conveyor belt at a rate proportional to the square root of speed (v) of the belt, i.e. $\frac{\mathrm{dm}}{\mathrm{dt}} \propto \sqrt{v}$. If P is the power delivered to run the belt at constant speed then which of the following relationship is true?

— Skipped

Correct Answer: C

Physics — Section B — Q7.
Three equal masses m are kept at vertices (A, B, C) of an equilateral triangle of side a in free space. At $t=0$, they are given an initial velocity $\vec{V}_A = V_0 AC$, $\vec{V}_B = V_0 BA$ and $\vec{V}_C = V_0 CB$. Here, $\overrightarrow{AC}, \overrightarrow{CB}$ and $\overrightarrow{BA}$ are unit vectors along the edges of the triangle. If the three masses interact gravitationally, then the magnitude of the net angular momentum of the system at the point of collision is:



— Skipped

Correct Answer: C

Physics — Section B — Q8. The number of spectral lines emitted by atomic hydrogen that is in the 4^{th} energy level, is

— Skipped

Correct Answer: C

Physics — Section B — Q9.
A convex lens made of glass (refractive index = 1.5) has focal length 24 cm in air. When it is totally immersed in water (refractive index = 1.33), its focal length changes to

— Skipped

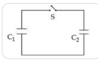
Correct Answer: B

Physics — Section B — Q10. In an experiment with photoelectric effect, the stopping potential,

— Skipped

Correct Answer: D

Physics — Section B — Q11.
A capacitor, $C_1 = 6\mu\mathrm{F}$ is charged to a potential difference of $\mathrm{V}_0 = 5\mathrm{V}$ using a $5\mathrm{V}$ battery. The battery is removed and another capacitor, $\mathrm{C}_2 = 12\mu\mathrm{F}$ is inserted in place of the battery. When the switch 'S' is closed, the charge flows between the capacitors for some time until equilibrium condition is reached. What are the



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Correct Answer: A

Physics — Section B — Q12.
A plane electromagnetic wave propagates along the +x direction in free space. The components of the electric field, $\vec{E}$ and magnetic field, $\vec{B}$ vectors associated with the wave in Cartesian frame are

— Skipped

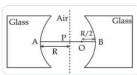
Correct Answer: B

Physics — Section B — Q13.
A cup of coffee cools from 90° C to 80° C in t minutes when the room temperature is 20° C. The time taken by the similar cup of coffee to cool from 80° C to 60° C at the same room temperature is :

— Skipped

Correct Answer: D

Physics — Section B — Q14.
Two concave refracting surfaces of equal radii of curvature and refractive index 1.5 face each other in air as shown in figure. A point object O is placed midway, between P and B. The separation between the images of O, formed by each refracting surface is :



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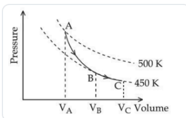
Correct Answer: D

Physics — Section B — Q15.
A poly-atomic molecule ($C_V = 3R, C_P = 4R$, where R is gas constant) goes from phase space point A ($\mathrm{P_A} = 10^5$ Pa, $\mathrm{V_A} = 4 \times 10^{-6}$ m^3) to point B ($\mathrm{P_B} = 5 \times 10^4$ Pa, $\mathrm{V_B} = 6 \times 10^{-6}$ m^3) to point



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Correct Answer: B

Physics — Section B — Q16.

Match List - I with List - II. List - I List - II (A) Magnetic induction (I) Ampere meter (B) Magnetic intensity (II) Weber (C) Magnetic flux (III) Gauss (D) Magnetic moment (IV) Ampere/meter

— Skipped

Correct Answer: D

Physics — Section B — Q17.

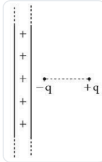
A point charge causes an electric flux of $-2 \times 10^4 Nm^2C^{-1}$ to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is: (Given $\epsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$)

— Skipped

Correct Answer: D

Physics — Section B — Q18.

An electric dipole is placed at a distance of 2 cm from an infinite plane sheet having positive charge density σ_0 . Choose the correct option from the following.



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Correct Answer: D

Physics — Section B — Q19.

The magnetic field inside a 200 turns solenoid of radius 10 cm is 2.9×10^{-4} Tesla. If the solenoid carries a current of 0.29 A, then the length of the solenoid is ____ π cm.

— Skipped

Correct Answer: 8

Physics — Section B — Q20.

A parallel plate capacitor consisting of two circular plates of radius 10 cm is being charged by a constant current of 0.15 A. If the rate of change of potential difference between the plates is 7×10^8 V/s then the integer value of the distance between the parallel plates is (Take, $\epsilon_0 = 9 \times 10^{-12} \frac{F}{m}, \pi = \frac{22}{7}$) ____ μm .

— Skipped

Correct Answer: 1320

Physics — Section B — Q21.

Two planets, A and B are orbiting a common star in circular orbits of radii R_A and R_B , respectively, with $R_B = 2R_A$. The planet B is $4\sqrt{2}$ times more massive than planet A. The ratio $\left(\frac{L_B}{L_A}\right)$ of angular momentum (L_B) of planet B to that of planet $A(L_A)$ is closest to integer ____.

— Skipped

Correct Answer: 8

Physics — Section B — Q22.

Two cars P and Q are moving on a road in the same direction. Acceleration of car P increases linearly with time whereas car Q moves with a constant acceleration. Both cars cross each other at time t = 0, for the first time. The maximum possible number of crossing(s) (including the crossing at t = 0) is ____.

— Skipped

Correct Answer: 3

Physics — Section B — Q23.

A physical quantity Q is related to four observables a, b, c, d as follows: $Q = \frac{ab^4}{cd}$ where, $a = (60 \pm 3)\text{Pa}$; $b = (20 \pm 0.1)\text{m}$; $c = (40 \pm 0.2)\text{Nsm}^{-2}$ and $d = (50 \pm 0.1)\text{m}$, then the percentage error in Q is $\frac{x}{1000}$, where $x =$ ____.

— Skipped

Correct Answer: 7700

Chemistry — Section A — Q1.

If the pressure applied over the system $CO(g) + 3H2(g) \rightleftharpoons CH4(g) + H2O(g)$ increases by two fold at constant temperature then (A) Concentration of reactants and products increases. (B) Equilibrium will shift in forward direction. (C) Equilibrium constant increases since concentration of products increases. (D) Equilibrium constant remains unchanged as concentration of reactants and products remain same. Choose the correct answer from the options given below :

— Skipped

Correct Answer: B

Chemistry — Section A — Q2.

Drug X becomes ineffective after 50% decomposition. The original concentration of drug in a bottle was 16mg/mL which becomes 4mg/mL in 12 months. The expiry time of the drug in months is ____. Assume that the decomposition of the drug follows first order kinetics.

— Skipped

Correct Answer: B

Chemistry — Section B — Q1. O_2 gas will be evolved as a product of electrolysis of:

— Skipped

Correct Answer: B

Chemistry — Section B — Q2. Identify the homoleptic complexes with odd number of d electrons in the central metal :

— Skipped

Correct Answer: C

Chemistry — Section B — Q3. Which one of the following, with HBr will give a phenol?

— Skipped

Correct Answer: C

Chemistry — Section B — Q4.

Given below are two statements: Statement (I): NaCl is added to the ice at 0° C, present in the ice cream box to prevent the melting of ice cream. Statement (II): On addition of NaCl to ice at 0° C, there is a depression in freezing point. In the light of the above statements, choose the correct answer from the options given below:

— Skipped

Correct Answer: D

Chemistry — Section B — Q5.

Match List - I with List - II : Choose the correct answer from the options given below :

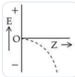
	List - I Applications		List - II Batteries/Cell
(A)	Transistors	(I)	Anode - Zn/Hg ; Cathode - HgO + C
(B)	Hearing aids	(II)	Hydrogen fuel cell
(C)	Invertors	(III)	Anode - Zn ; Cathode - Carbon
(D)	Apollo space ship	(IV)	Anode - Pb ; Cathode - Pb | PbO_2

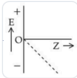
— Skipped

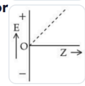
Correct Answer: C

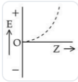
Chemistry — Section B — Q6.

For hydrogen like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n ? [E: Energy of the stationary state, Z : atomic number, n = principal quantum number]









— Skipped

Correct Answer: A

Chemistry — Section B — Q7. Which one of the following reaction sequences will give an azo dye?

— Skipped

Correct Answer: A

Chemistry — Section B — Q8. Identify the essential amino acids from below :

— Skipped

Correct Answer: B

Chemistry — Section B — Q9. The calculated spin-only magnetic moments of K_3 [Fe(OH)$_6$] and K_4 [Fe(OH)$_6$] respectively are :

— Skipped

Correct Answer: D

Chemistry — Section B — Q10.

The type of oxide formed by the element among Li, Na, Be, Mg, B and Al that has the least atomic radius is :

— Skipped

Correct Answer: B

Chemistry — Section B — Q11. Total number of sigma (σ) and pi(π) bonds respectively present in hex-1-en-4-yne are :

— Skipped

Correct Answer: C

Chemistry — Section B — Q12.

First ionisation enthalpy values of first four group 15 elements are given below. Choose the correct value for the element that is a main component of apatite family :

— Skipped

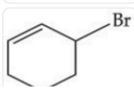
Correct Answer: C

Chemistry — Section B — Q13.

Given below are two statements: Statement (I): On nitration of m-xylene with HNO_3 , H_2SO_4 followed by oxidation, 4-nitrobenzene-1,3-dicarboxylic acid is obtained as the major product. Statement (II): $-CH_3$ group is o/p-directing while $-NO_2$ group is m-directing group. In the light of the above statements, choose the correct answer from the options given below:

— Skipped

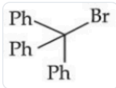
Correct Answer: D

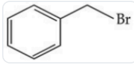


Chemistry — Section B — Q14

Chemistry — Section B — Q14.

Which among the following halides will generate the most stable carbocation in the nucleophilic substitution reaction?





— Skipped

Correct Answer: C

Chemistry — Section B — Q15.

Given below are two statements: Statement (I): It is impossible to specify simultaneously with arbitrary precision, both the linear momentum and the position of a particle. Statement (II): If the uncertainty in the measurement of position and uncertainty in measurement of momentum are equal for an electron, then the uncertainty in the measurement of velocity is $\geqslant \sqrt{\frac{h}{\pi}} \times \frac{1}{2m}$. In the light of the above statements, choose the correct answer from the options given below:

— Skipped

Correct Answer: C

Chemistry — Section B — Q16.

0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solutions. According to equation $5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$ Identify the correct statements:

— Skipped

Correct Answer: A

Chemistry — Section B — Q17.

Given below are two statements: Statement (I): In partition chromatography, stationary phase is thin film of liquid present in the inert support. Statement (II): In paper chromatography, the material of paper acts as a stationary phase. In the light of the above statements, choose the correct answer from the options given below:

— Skipped

Correct Answer: A

Chemistry — Section B — Q18.

If C (diamond) $\rightarrow C$ (graphite) + XkJmol^{-1} C (diamond) + O_2 (g) $\rightarrow$ CO_2 (g) + YkJmol^{-1} C (graphite) + O_2 (g) $\rightarrow$ CO_2 (g) + ZkJmol $^{-1}$ at constant temperature. Then

— Skipped

Correct Answer: D

Chemistry — Section B — Q19.

In the sulphur estimation, 0.20 g of a pure organic compound gave 0.40 g of barium sulphate. The percentage of sulphur in the compound is ____ ×10^{-1}%. (Molar mass : O = 16, S = 32, Ba = 137 in gmol^{-1})

— Skipped

Correct Answer: 275

Chemistry — Section B — Q20.

Isomeric hydrocarbons $\rightarrow$ negative Baeyer's test (Molecular formula C_9H_{12}) The total number of isomers from above with four different non-aliphatic substitution sites is -

— Skipped

Correct Answer: 2

Chemistry — Section B — Q21.

Consider the following low-spin complexes $K_3[Co(NO_2)_6]$, $K_4[Fe(CN)_6]$, $K_3[Fe(CN)_6]$, $Cuu_2[Fe(CN)_6]$ and $Zn_2[Fe(CN)_6]$. The sum of the spin-only magnetic moment values of complexes having yellow colour is. ____ B.M. (answer in nearest integer)

— Skipped

Correct Answer: 0

Chemistry — Section B — Q22.

In the Claisen-Schmidt reaction to prepare, dibenzalacetone from 5.3 g of benzaldehyde, a total of 3.51 g of product was obtained. The percentage yield in this reaction was ____%.

— Skipped

Correct Answer: 60

Chemistry — Section B — Q23. Total number of non bonded electrons present in ion based on Lewis theory is

— Skipped

Correct Answer: 12

Mathematics — Section A — Q1. Let $f(x) = \int_0^t t(t^2 - 9t + 20)dt, 1 \leq x \leq 5$. If the range of f is $[\alpha, \beta]$, then $4(\alpha + \beta)$ equals:

— Skipped

Correct Answer: D

Mathematics — Section A — Q2.

Let $\hat{a}$ be a unit vector perpendicular to the vectors $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} - \hat{k}$, and makes an angle of $\cos^{-1}\left(-\frac{1}{3}\right)$ with the vector $\hat{i} + \hat{j} + \hat{k}$. If $\hat{a}$ makes an angle of $\frac{\pi}{3}$ with the vector $\hat{i} + \alpha\hat{j} + \hat{k}$, then the value of α is:

— Skipped

Correct Answer: B

Mathematics — Section B — Q1.

If for the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + (\tan x)y = \frac{2+\sec x}{(1+2\sec x)^2}$, $x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$, $f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{10}$, then $f\left(\frac{\pi}{4}\right)$ is equal to:

— Skipped

Correct Answer: D

Mathematics — Section B — Q2.
Let P be the foot of the perpendicular from the point $(1,2,2)$ on the line L: $\frac{x-1}{1}=\frac{y+1}{-1}=\frac{z-2}{2}$. Let the line $\vec{r}=(-\hat{i}+\hat{j}-2\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k}), \lambda \in \mathbf{R}$, intersect the line L at Q. Then 2(PQ) ^{2} is equal to :

— Skipped

Correct Answer: D

Mathematics — Section B — Q3.
Let $A=[a_{ij}]$ be a matrix of order 3×3, with $a_{ij}=(\sqrt{2})^{i+j}$. If the sum of all the elements in the third row of A^2 is $\alpha+\beta\sqrt{2}, \alpha, \beta \in \mathbf{Z}$, then $\alpha+\beta$ is equal to:

— Skipped

Correct Answer: B

Mathematics — Section B — Q4.
Let the line $x+y=1$ meet the axes of x and y at A and B, respectively. A right angled triangle AMN is inscribed in the triangle OAB, where O is the origin and the points M and N lie on the lines OB and AB, respectively. If the area of the triangle AMN is $\frac{4}{9}$ of the area of the triangle OAB and $AN:NB=\lambda:1$, then the sum of all possible value(s) of is λ:

— Skipped

Correct Answer: A

Mathematics — Section B — Q5.
If all the words with or without meaning made using all the letters of the word "KANPUR" are arranged as in a dictionary, then the word at 440^{th} position in this arrangement, is :

— Skipped

Correct Answer: C

Mathematics — Section B — Q6.
If the set of all a $\in \mathbf{R}$, for which the equation $2x^2+(a-5)x+15=3$a has no real root, is the interval (α,β), and $X=\{x\in Z: \alpha<x<\beta\}$, then $\sum_{x\in X} x^2$ is equal to:

— Skipped

Correct Answer: D

Mathematics — Section B — Q7.
Let $A=[\mathrm{a}_{ij}]$ be a 2×2 matrix such that $\mathrm{a}_{ij}\in\{0,1\}$ for all i and j. Let the random variable X denote the possible values of the determinant of the matrix A. Then, the variance of X is :

— Skipped

Correct Answer: C

Mathematics — Section B — Q8.
Let the function $f(x)=(x^2+1)|x^2-ax+2|+\cos|x|$ be not differentiable at the two points $x=\alpha=2$ and $x=\beta$. Then the distance of the point (α,β) from the line $12x+5y+10=0$ is equal to :

— Skipped

Correct Answer: C

Mathematics — Section B — Q9.
Let the area enclosed between the curves $|y|=1-x^2$ and $x^2+y^2=1$ be α. If $9\alpha=\beta\pi+\gamma; \beta,\gamma$ are integers, then the value of $|\beta-\gamma|$ equals.

— Skipped

Correct Answer: B

Mathematics — Section B — Q10. The remainder, when 7^{103} is divided by 23, is equal to :

— Skipped

Correct Answer: D

Mathematics — Section B — Q11.
If $\alpha x+\beta y=109$ is the equation of the chord of the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, whose mid point is $\left(\frac{5}{2},\frac{1}{2}\right)$, then $\alpha+\beta$ is equal to:

— Skipped

Correct Answer: A

Mathematics — Section B — Q12.
If the domain of the function $\log_5(18x-x^2-77)$ is (α,β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ,δ), then $\alpha^2+\beta^2+\gamma^2$ is equal to:

— Skipped

Correct Answer: C

Mathematics — Section B — Q13.
Let a circle C pass through the points (4, 2) and (0, 2), and its centre lie on $3x+2y+2=0$. Then the length of the chord, of the circle C, whose mid-point is (1, 2), is:

— Skipped

Correct Answer: C

Mathematics — Section B — Q14.
Let a straight line L pass through the point $P(2,-1,3)$ and be perpendicular to the lines $\frac{x-1}{2}=\frac{y+1}{1}=\frac{z-3}{-2}$ and $\frac{x-3}{1}=\frac{y-2}{3}=\frac{z+2}{4}$. If the line L intersects the yz-plane at the point Q, then the distance between the points P and Q is:

— Skipped

Correct Answer: D

Mathematics — Section B — Q15.
Bag 1 contains 4 white balls and 5 black balls, and Bag 2 contains n white balls and 3 black balls. One ball is drawn randomly from Bag 1 and transferred to Bag 2. A ball is then drawn randomly from Bag 2. If the probability, that the ball drawn is white, is $29/45$, then n is equal to :

— Skipped

Correct Answer: A

Mathematics — Section B — Q16.
Let $\alpha, \beta (\alpha \neq \beta)$ be the values of m, for which the equations $x + y + z = 1; x + 2y + 4z = m$ and $x + 4y + 10z = m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10}(n^\alpha + n^\beta)$ is equal to:

— Skipped

Correct Answer: D

Mathematics — Section B — Q17.
Let $S = \mathrm{N} \cup \{0\}$. Define a relation R from S to $\mathbf{R}$ by: $\mathbf{R} = \left\{(x, y) : \log_e y = x \log_e \left(\frac{2}{5}\right), x \in \mathrm{S}, y \in \mathbf{R}\right\}$. Then, the sum of all the elements in the range of R is equal to:

— Skipped

Correct Answer: D

Mathematics — Section B — Q18.
If $\sin x + \sin^2 x = 1, x \in (0, \frac{\pi}{2})$, then $\left(\cos^{12} x + \tan^{12} x\right) + 3\left(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x\right) + \left(\cos^6 x + \tan^6 x\right)$ is equal to :

— Skipped

Correct Answer: D

Mathematics — Section B — Q19.
If $24 \int_0^{\frac{\pi}{4}} \left(\sin\left|4x - \frac{\pi}{12}\right| + [2\sin x]\right) dx = 2\pi + \alpha$, where $[\cdot]$ denotes the greatest integer function, then α is equal to ____.

— Skipped

Correct Answer: 12

Mathematics — Section B — Q20.
Let $a_1, a_2, \ldots, a_{2024}$ be an Arithmetic Progression such that $a_1 + (a_5 + a_{10} + a_{15} + \ldots + a_{2020}) + a_{2024} = 2233$. Then $a_1 + a_2 + a_3 + \ldots + a_{2024}$ is equal to

— Skipped

Correct Answer: 11132

Mathematics — Section B — Q21. If $\lim_{t \to 0} \left(\int_0^1 (3x+5)^t \mathrm{d}x\right)^{\frac{1}{t}} = \frac{\alpha}{5\mathrm{e}} \left(\frac{8}{5}\right)^{\frac{2}{3}}$, then α is equal to

— Skipped

Correct Answer: 64

Mathematics — Section B — Q22.
Let $y^2 = 12x$ be the parabola and S be its focus. Let PQ be a focal chord of the parabola such that (SP)(SQ) = $\frac{147}{4}$. Let C be the circle described taking PQ as a diameter. If the equation of a circle C is $64x^2 + 64y^2 - \alpha x - 64\sqrt{3}y = \beta$, then $\beta - \alpha$ is equal to ____.

— Skipped

Correct Answer: 1328

Mathematics — Section B — Q23.
Let integers $a, b \in [-3, 3]$ be such that $a + b \neq 0$. Then the number of all possible ordered pairs (a, b), for which $\left|\frac{z-a}{z+b}\right| = 1$ and $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1, z \in \mathbb{C}$, where ω and ω^2 are the roots of $x^2 + x + 1 = 0$, is equal to ____.

— Skipped

Correct Answer: 10

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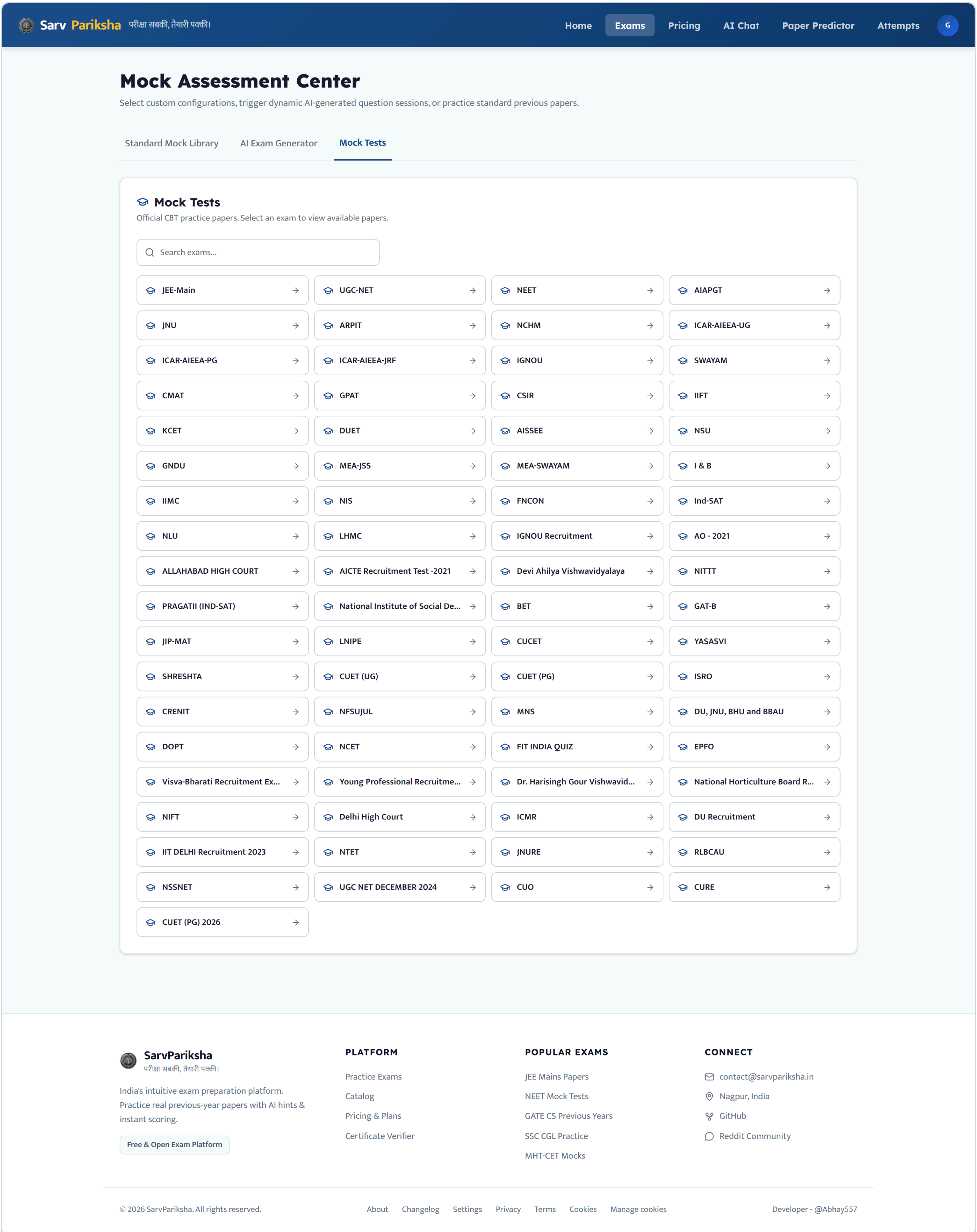

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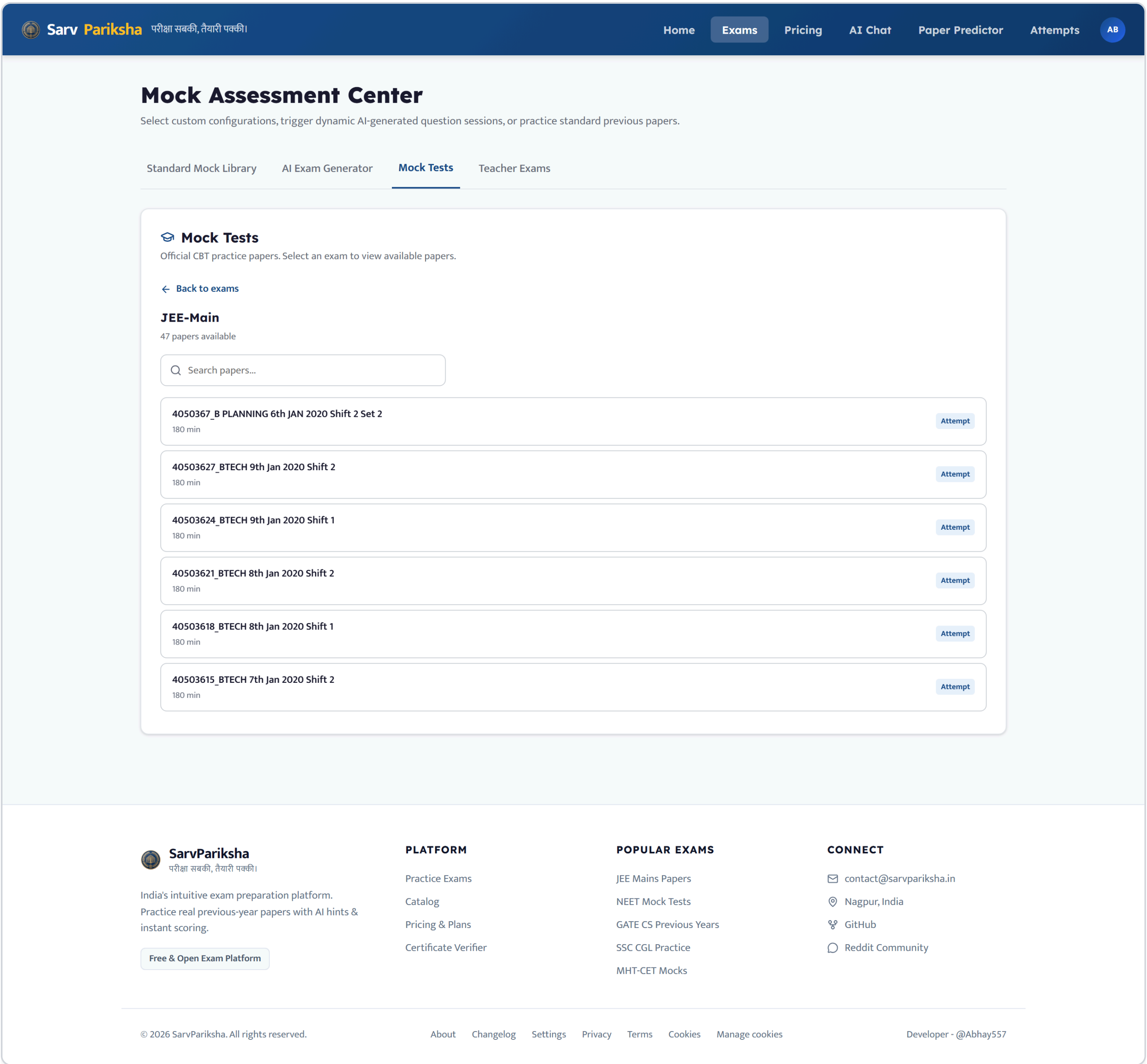

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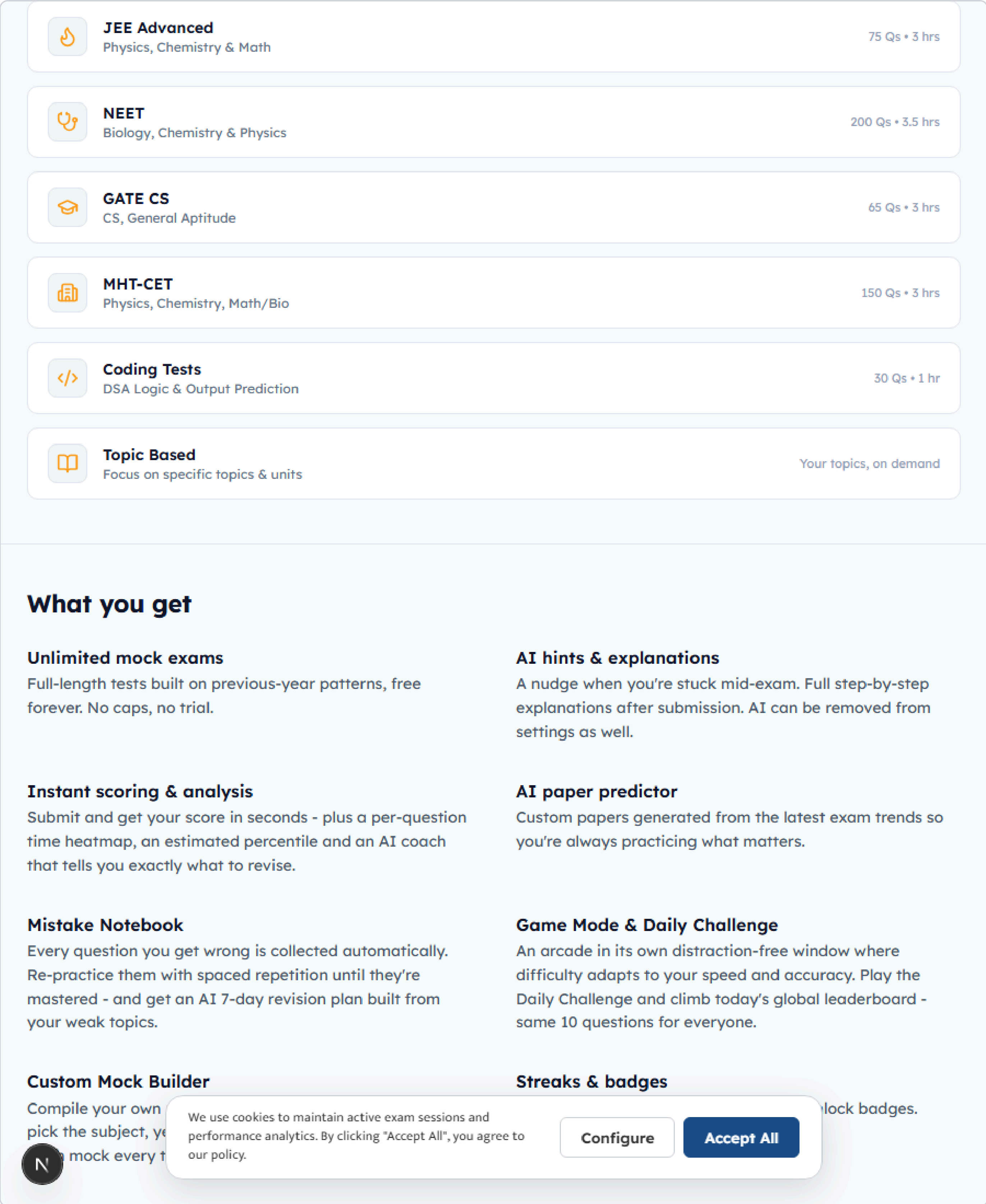

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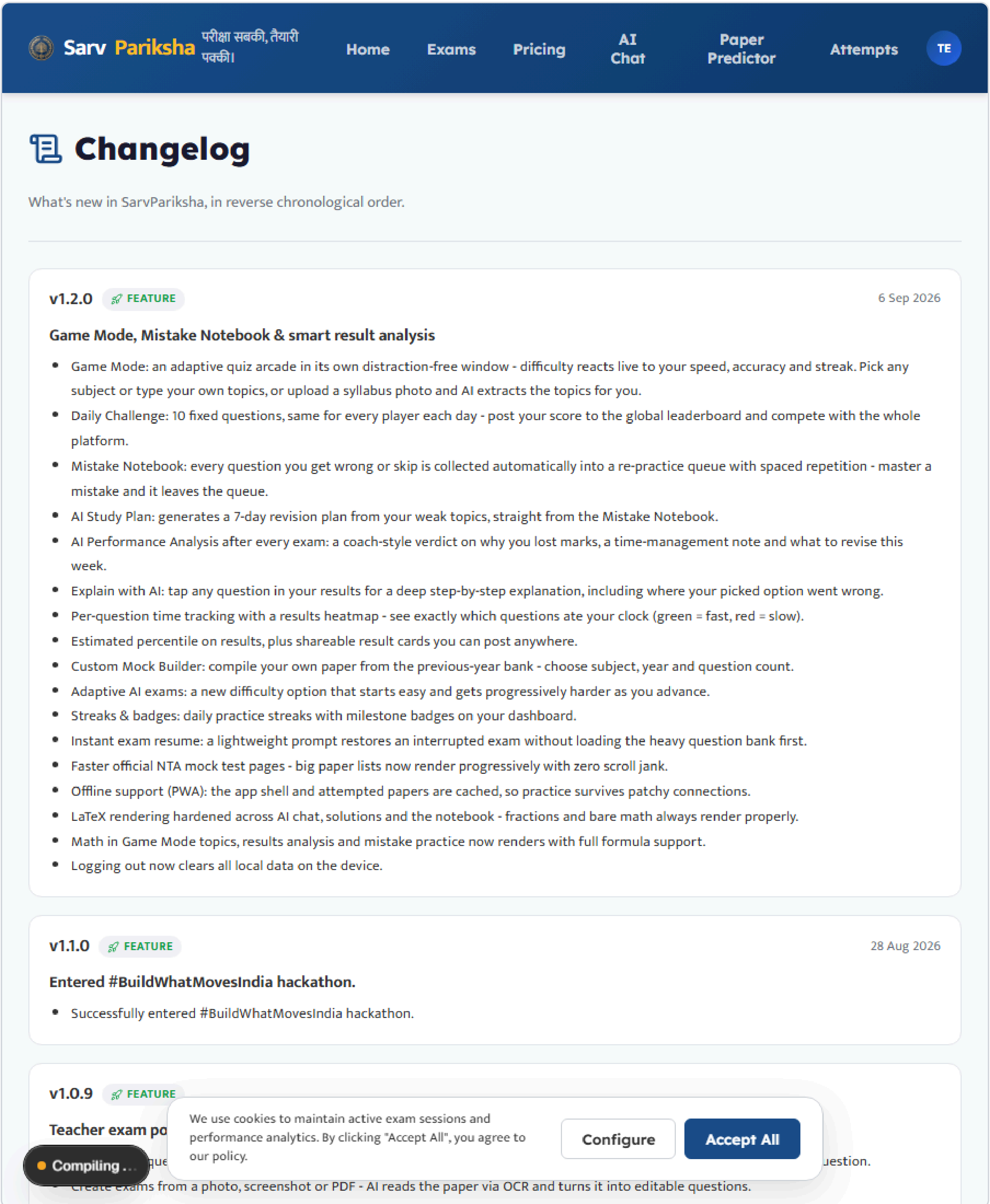

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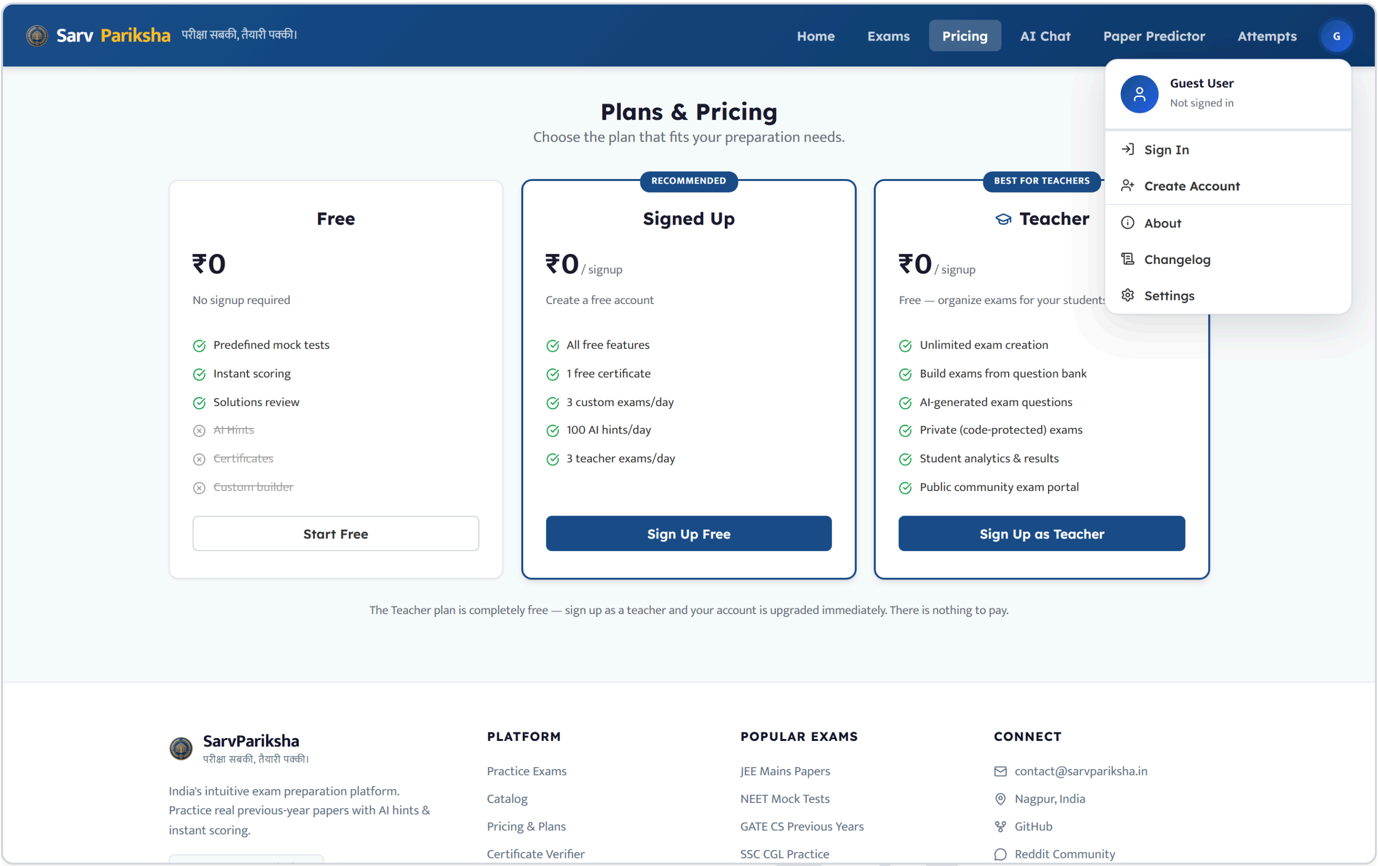

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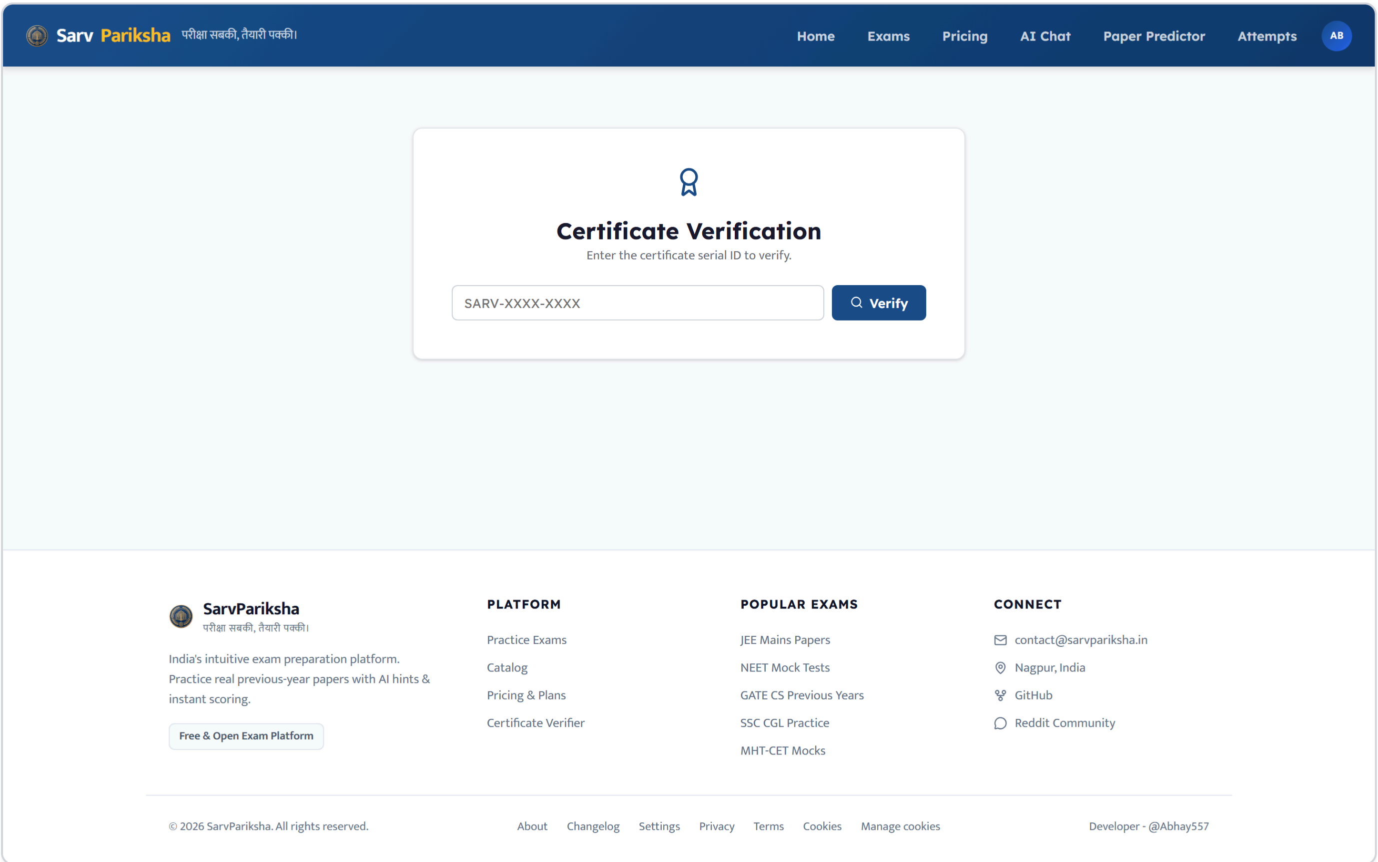

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